

Redrob Data & AI Challenge

Next-Gen Candidate Screening System

Team Leader: Shreya Gupta

Building a blazing-fast, accurate, and completely offline system that runs on standard hardware while explaining exactly why a candidate was chosen.

| Lightweight NLP Pipeline



Zero Hallucinations

No LLMs for reasoning. Our explainability is deterministic, pulling actual matching skills from candidate data.



Skill Boosting

Explicit skills are doubled in weight, prioritizing verified expertise over general summary "fluff".



CPU Optimized

Optimized matrix math instead of heavy neural networks. Runs locally on standard hardware.

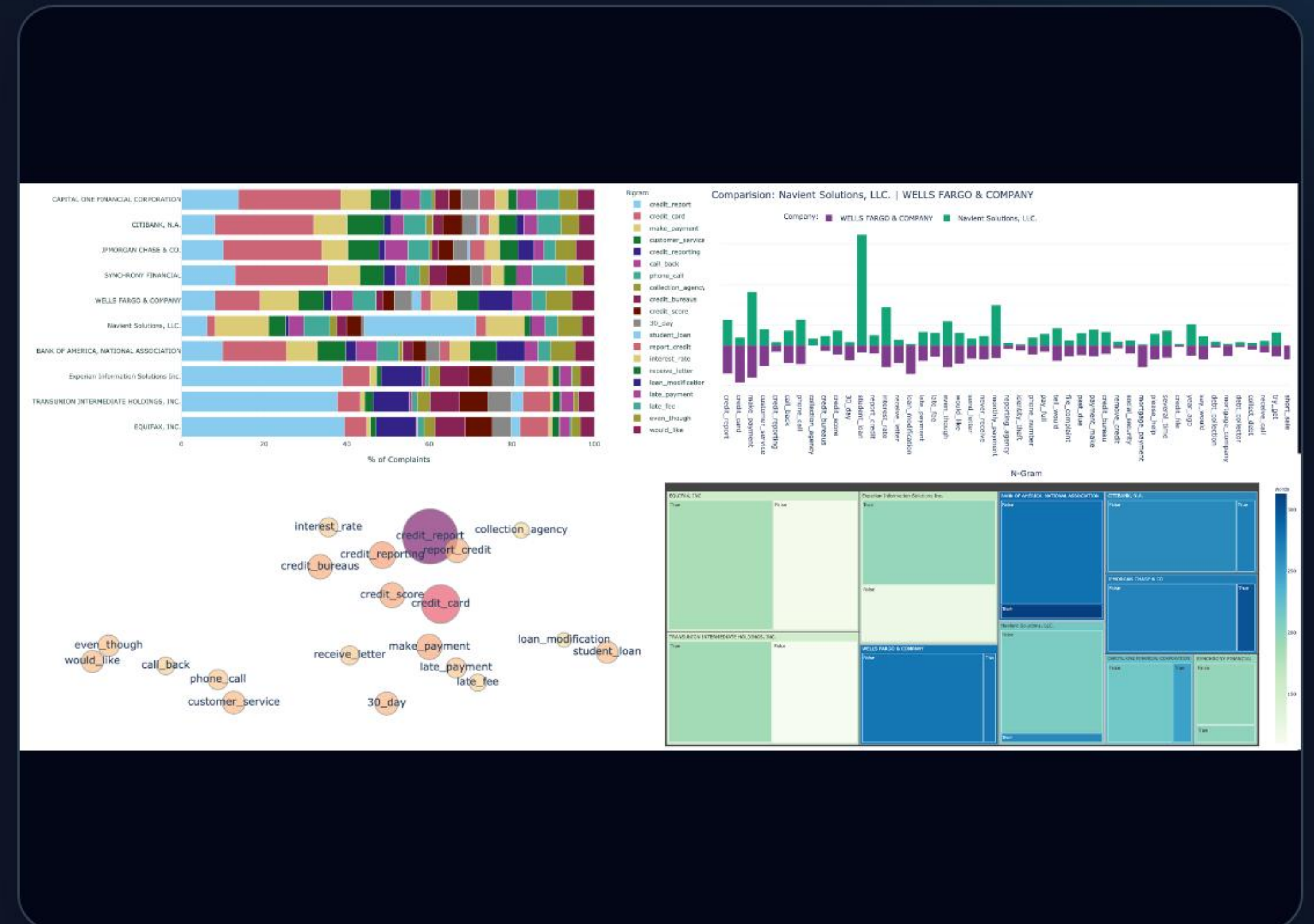
Evaluation & Signals

JD Understanding

- ✓ Dynamic extraction of vocabulary (3+ chars).
- ✓ Core "keyword foundation" construction.
- ✓ Double weight given to explicit skill tags.

Beyond Keywords

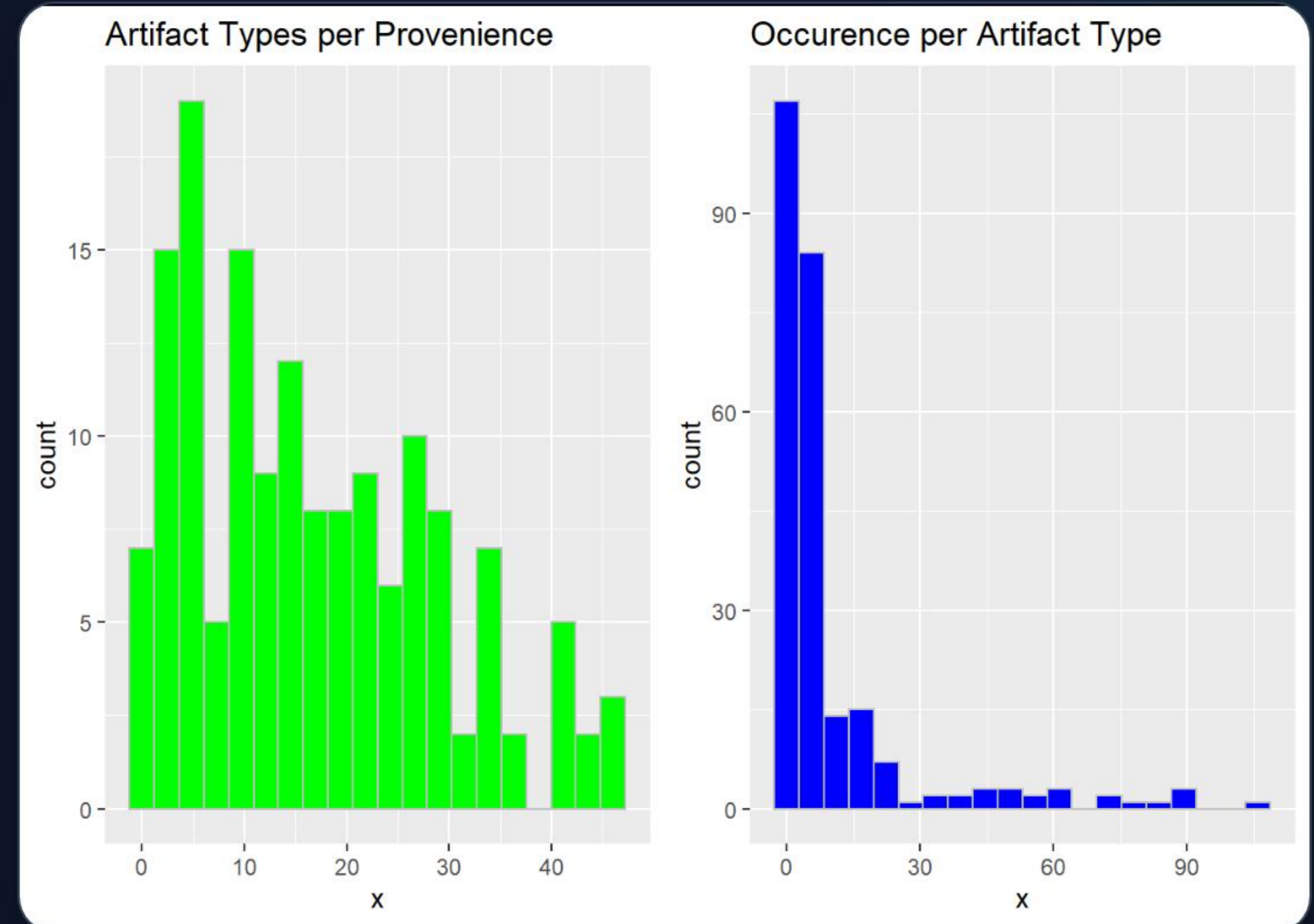
Utilizes **Bi-Grams** to understand context. Recognizes "Machine Learning" as a single skill phrase.



Ranking Methodology

Models & Algorithms

- ✓ **TF-IDF Vectorization:** Capped at 20,000 features for extreme throughput.
- ✓ **Cosine Similarity:** Calculates precise semantic overlap scores (0.0 - 1.0).
- ✓ **Deterministic Ranking:** Tied scores are broken using unique Candidate IDs.



| Explainability & Validation

Reasoning Engine

Generates factual justifications calling out exact Years of Experience and top 3 direct skill matches.

Hallucination Check

Rule-based formatting ensures the system only outputs skills genuinely found in JSON profiles.

Quality Filtering

Low-quality or suspicious profiles naturally fall to the bottom due to low semantic overlap.

End-to-End Workflow

1. Ingestion

Load JD text and
parse JSONL data.

2. Extraction

Pull headlines &
boost skills by 2x.

3. Vectorization

TF-IDF Bi-Gram
matrix conversion.

4. Scoring

Calculate Cosine
Similarity.

5. Ranking

Sort & isolate Top
100 candidates.

6. Reasoning

Auto-generate
factual justifications.

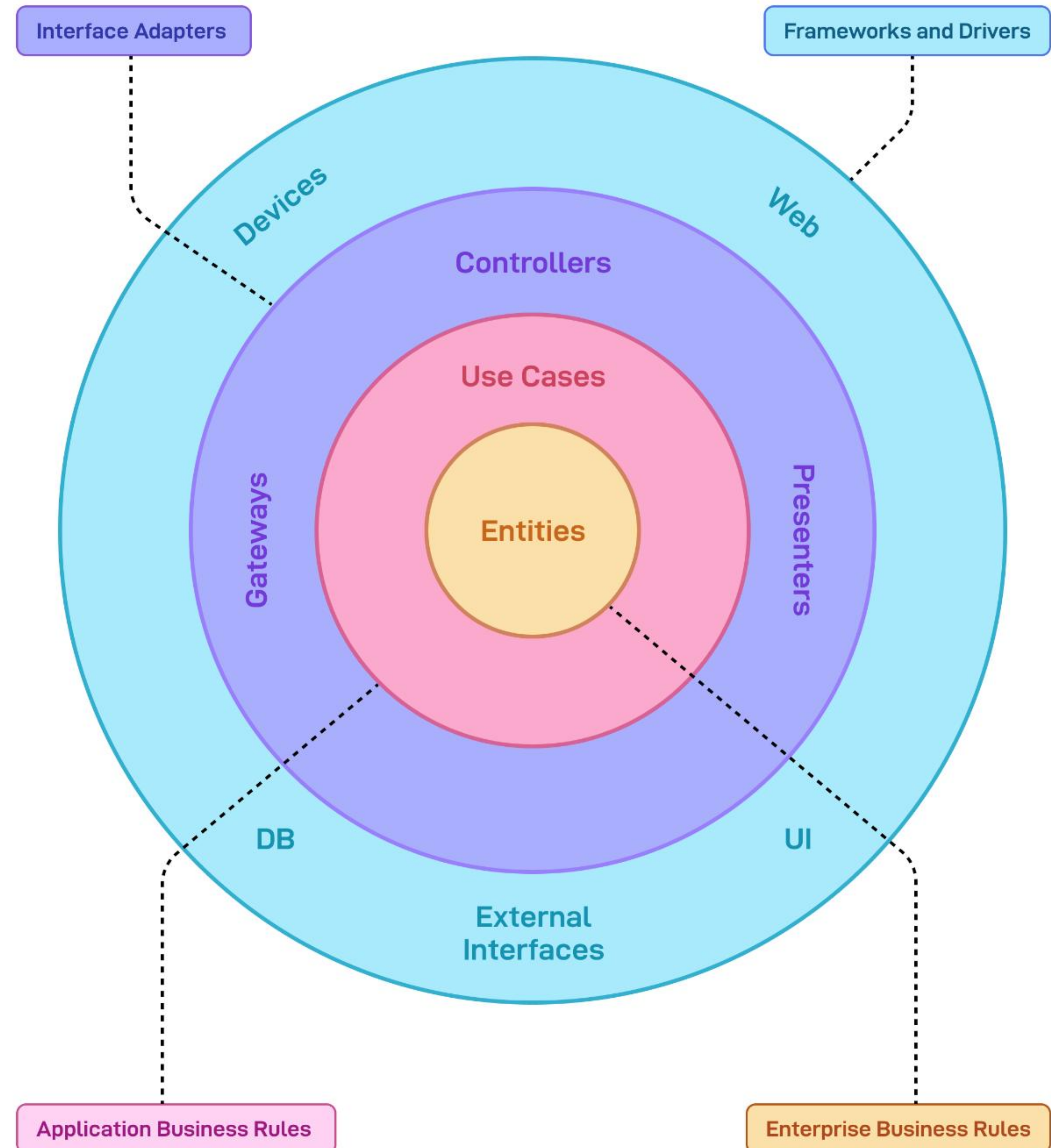
7. Export

Save ranked results
to clean CSV.

System Architecture

Logical Layers

- ✓ **Data Layer:** job_description.txt + candidates.jsonl
- ✓ **Processing:** Python & Pandas Data Cleaning
- ✓ **NLP Engine:** Scikit-Learn Tfidf + Cosine Similarity
- ✓ **Explainability:** Rule-based logic module
- ✓ **Output:** Final Ranked CSV (team_123.csv)

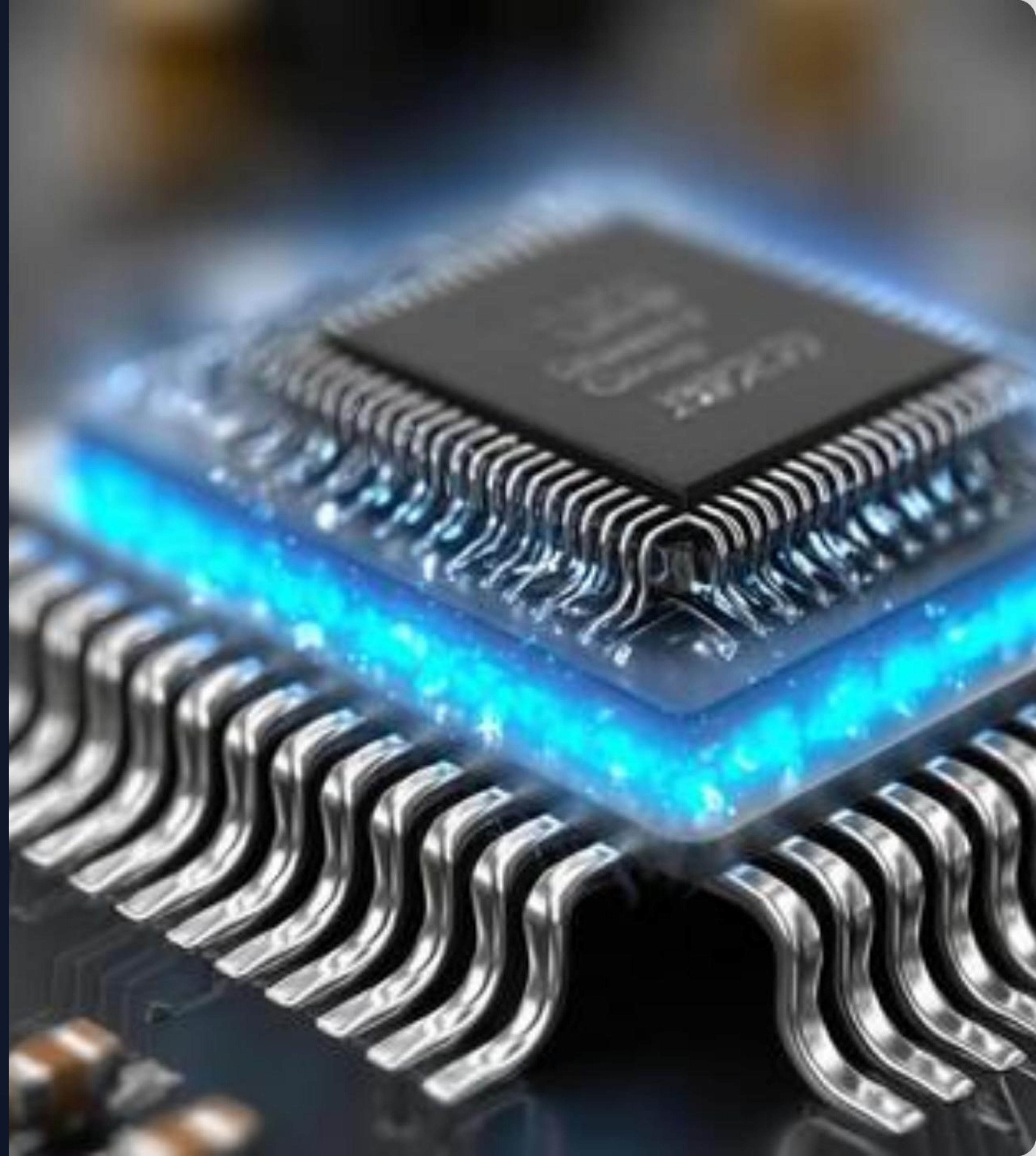


| Results & Performance

Ranking Quality: Capturing two-word phrases effectively ignores "fluff" and surfaces candidates with genuine technical capabilities.

Compute Constraints: Relies on highly optimized C-backend operations (Scikit-Learn). Processes thousands of profiles in seconds on standard CPUs.

- ✓ Zero Cloud Compute Costs.
- ✓ Zero GPU Required.
- ✓ Full Offline Execution.



| Technologies Used



Python

Core logic and readable data handling.



Pandas

Blazing-fast data manipulation & sorting.



Scikit-Learn

Industry standard for vector math and similarity.



JSON & Regex

Dependency-free text parsing and cleaning.

Submission Assets

GitHub Repository & Video Demo Links

 **GitHub Repo:**

github.com/Shreya-Gupta-5074/Redrob-Data-AI-Challenge

 **Video Demo:**

[Google Drive Demo Link](#)

Redrob AI Screening Challenge Submission

Image Sources



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